MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

RETAINING RINGS

TO ALL HOLDERS OF DOD-STD-35-40A(MI):

1. THE FOLLOWING PAGES OF DOD-STD-35-40A(MI) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
1	28 AUGUST 1978	(Reprinted without change)	
2	30 JUNE 1992	2	28 AUGUST 1978
3	30 JUNE 1992	3	28 AUGUST 1978
4	30 JUNE 1992	4	28 AUGUST 1978
33	28 AUGUST 1978	(Reprinted without change)	
34	30 JUNE 1992	34	28 AUGUST 1978
35	30 JUNE 1992	35	28 AUGUST 1978
36	28 AUGUST 1978	(Reprinted without change)	

2. RETAIN THIS NOTICE PAGE AND INSERT BEFORE THE TABLE OF CONTENTS.

3. Holders of DOD-STD-35-40A(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
Army-MI

Review activities:
Army-AT
Army-AV
Army-ER
Army-ME
Army-MR
Army-SC
Army-CR

User activity:
Army - AR

Preparing activity:
Army - MI

(Project EGDS-A416)

AREA EGDS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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1. SCOPE

1.1 General. This dash-numbered part of the Automated Engineering Document Preparation System (AEDPS), provides detailed requirements for the preparation of Military Specification Exception (MSE) documentation for retaining rings.

1.2 Purpose. This military standard provides the necessary information to prepare a Specification Requirement Sheet (SRS) which will be used to generate an MSE by automated data processing.

1.3 Utilization. General instructions, quality assurance provisions, packaging, and notes, applicable to all AEDPS dash-numbered parts have been included in the basic document (DOD-STD-35) to avoid redundancy. Unless otherwise specified in this dash-numbered part, the following list contains additional subjects that are applicable to all AEDPS dash-numbered parts and are included in DOD-STD-35:

a. Writing Methodology
b. Document and Part Numbering
c. Applicable Documents
d. Single and Multi-item Specifications (Tabulation)
e. Keyword Code Index (KCI)
f. General Requirements of AEDPS

1.4 Military Specification Exception Title Index. The title code for an MSE shall be selected from one of the MSE titles listed in the Title Index (see table I). Data bases have been established for the generation of MSE°s for the MSE titles indicated. If a suitable title code is not contained in the Title Index, leave the title code spaces on the SRS blank, and attach a request for the assignment of a title code. Include in the request the desired title and the base document identifier. A code will be assigned and the code, title, and base document will be made a part of the Title Index.

One of the six-digit codes listed in the Title Index entered on an SRS, will specify that the title of the MSE will be the title shown in the Title Index.

Each MSE title listed in the Title Index is compatible with the document(s) listed with that title under base document. The list of base documents may or may not be a complete listing. The information shown under coverage is intended for the user°s reference; when more than one document is listed for an MSE title, this information will be found useful in determining which of the documents is most suitable for use as the base document.

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The MSE title may or may not be the same as the title used in DODISS, FIIG, or the base document. In general, the MSE title uses the basic federal item name listed in Cataloging Handbook H6. However, titles with additional modifying and identifying words are provided to assist in defining the scope of the MSE. Also, less restrictive titles are provided to prevent exclusion of items.

TABLE I. TITLE INDEX

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
RINGS, RETAINING, REDUCED SECTION	161001	MIL-R-21248	REDUCED SECTION
		MS3215	EXT, "E", REINF
		MS3216	EXT, PRONG-LOCK
		MS16632	EXT, CRESCENT
		MS16633	EXT, "E"
		MS16634	EXT, BOWED "E"
RINGS, RETAINING, TAPERED SECTION	161002	MIL-R-21248	TAPERED SECTION
		MS3217	EXT, HEAVY DUTY
		MS16624	EXT, BASIC
		MS16625	INT, BASIC
		MS16626	EXT, INVERTED
		MS16627	INT, INVERTED
		MS16628	EXT, BOWED
		MS16629	INT, BOWED
		MS16630	EXT, BEVELED
		MS16631	INT, BEVELED
		MS90707	EXT GRIP
		MS90708	EXT INTLK
RINGS, RETAINING, SPIRAL	161003	MIL-R-27426	SPIRAL RTNG RINGS
RINGS, RETAINING	161004	MIL-R-21248	REDUCED AND TAPERED SECTION
		MIL-R-27426	SPIRAL
		MS3211	SNAP RTNG RING
		MS5009	SNAP RTNG RING
		MS25086	LOCK RTNG RING
		MS51990	LOCK, SERRATED
		MS51997	LOCK, SERRATED, HIGH STRENGTH
RING, RETAINING, SCREW	161005	MIL-R-48211	EXTERNAL THREAD

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
GGG-P-480	Pliers, Retaining Ring

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MIL-P-16232	Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals)
MIL-R-21248	Rings, Retaining (Tapered and Reduced Section Type)
MIL-R-27426	Rings, Retaining, Spiral (Uniform Cross Section)
MIL-C-81562	Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited)

STANDARDS

MILITARY

DOD-STD-35	Automated Engineering Document Preparation Systems
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1312	Fasteners, Test Methods

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTMB633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTME10	Brinell Hardness of Metallic Materials
ASTME18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTME380	Metric Practice Guide

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

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CHAR CODE	STANDARD PARAGRAPH
1603	DISH. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1603.
1604	DISH. THE DISH SHALL BE NOT GREATER THAN 1604 INCH.
1606	DISH. THE DISH SHALL BE NOT GREATER THAN 1606 MILLIMETER(S).
1599	DISH. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1599.
1600	DISH. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH MIL-R-21248.
1602	DISH. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH MIL-R-27426.
1828	EMBRITTLEMENT RELIEF. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1828.
1830	EMBRITTLEMENT RELIEF. THE RETAINING RING SHALL NOT CRACK OR FAIL BY FRACTURE.
1802	ENBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1802.
1804	EMBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH QQ-P-416.

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
EMBRITTLEMENT RELIEF (CONT)	
TEST METHODS (CONT)	
PER ASTMB633.	1806
PER MIL-P-16232.	1808
PER MIL-C-81562.	1810
HARDNESS	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1862
HARDNESS (NUMBER)	1864
TEST INSTRUMENT (NAME)	1867
SCALE (LETTER)	1868
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1846
PER MIL-STD-1312, TEST 6.	1848
SURFACE PREPARATION BEFORE TEST ____	1849
SURFACE TO BE TESTED ____	1851
PER ASTME10.	1850

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CHAR CODE	STANDARD PARAGRAPH
1806	EMBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH ASTMB633.
1808	EMBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH MIL-P-16232.
1810	EMBRITTLEMENT RELIEF. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH MIL-C-81562.
1862	HARDNESS. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1862.
1864	HARDNESS. THE HARDNESS SHALL BE 1864.
1867	THE TEST INSTRUMENT SHALL BE 1867.
1868	THE 1868 SCALE SHALL APPLY.
1846	HARDNESS. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 1846.
1848	HARDNESS. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH TEST 6 OF MIL-STD-1312.
1849	SURFACE PREPARATION BEFORE TEST SHALL BE 1849.
1851	THE SURFACE TO BE TESTED SHALL BE 1851.
1850	HARDNESS. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH ASTME10.

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
HARDNESS (CONT)	
PER ASTME10. (CONT)	
SURFACE PREPARATION BEFORE TEST ____	1852
SURFACE TO BE TESTED ____	1853
PER ASTME18.	1854
SURFACE PREPARATION BEFORE TEST ____	1855
SURFACE TO BE TESTED ____	1856
HUMIDITY	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	4798
NO EVIDENCE OF CORROSION OR PHYSICAL DAMAGE.	4799
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	4788
PER MIL-STD-1312, TEST 3.	4881
RELATIVE HUMIDITY (PERCENT)	4789
DURATION OF TEST (HOURS)	4816
MISALIGNMENT	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1661

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